
What Survives the Filter? Trivial Baselines on Stored Honest/Attack Pairs in AI-Control Monitoring

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Abstract

1 Trusted monitoring in AI control is evaluated on stored honest/attack pairs. The
2 field already knows these pairs leak: it filters to working solutions and strips
3 comments. We ask what that preprocessing leaves behind. On the field’s own
4 filtered, comment-stripped split, a *trained* bag-of-words classifier scores above
5 roughly four-fifths of the open-weight monitors ranked on it, and a classifier
6 that reads no code at all — only problem metadata — scores above most of
7 them. We trace the residue to authorship rather than sabotage: the honest class
8 is human-written while every attack is model-written. The signal transfers across
9 backdoor generators, is nearly as strong on inert backdoors as on working ones,
10 and survives the field’s comment stripper. Regenerating only the honest half does
11 not isolate content: with an open-weight model writing the honest class against the
12 stored backdoors, separability *rises*. Holding authorship constant by construction,
13 with one model writing both classes, removes between a third (Claude Sonnet 5,
14 $\rho = 0.687$) and nine-tenths (Claude Haiku 4.5, $\rho = 0.095$) of the separability;
15 matching the attack prompt’s framing removes most of what remains, and what
16 survives on attacks that actually fire is consistent with surface content a monitor also
17 reads. A zero-shot monitor is inflated the same way, scoring 0.944 on stored pairs
18 and 0.818 on firing backdoors from the same author. We recommend that stored-
19 pair evaluations report partial-input and bag-of-words floors against permutation
20 nulls, use same-generator honest samples and the same prompt, and drop inert
21 backdoors from both classes.

22 1 Introduction

23 Trusted monitoring is the backbone protocol of AI control: a weaker trusted model reads an untrusted
24 model’s output and flags sabotage [4]. It is developed, tuned and ranked almost entirely on *stored*
25 honest/attack pairs — fixed corpora of solutions, some carrying a planted backdoor [12, 19, 9]. The
26 field already knows these pairs leak. Koran et al. keep only working honest solutions and working
27 backdoors and strip every comment, with the rationale stated [12]; Control Tax reports the pass-tests
28 pre-filter and appendices “Monitoring failed backdoors” [19]; [18] is an entire paper on the comment
29 channel. The sharpest treatment is Jha’s, whose harness ships a `-leak-check` flag with two trivial
30 baselines inside it and concludes that stripping is mandatory rather than cosmetic: a `#-count` detector
31 falls from 0.6957 to 0.4983 once comments go, and code length alone sits at 0.4726 [9]. This paper
32 starts from that work.

33 What nobody has measured is what the preprocessing leaves behind. On the field’s own filtered,
34 comment-stripped split, a bag-of-words classifier reaches AUROC 0.806 — above 20 of the 24
35 open-weight monitors ranked on that same split, whose median is 0.609 — and a classifier that reads
36 no code at all, only problem difficulty, test count and question length, reaches 0.772, above 20.

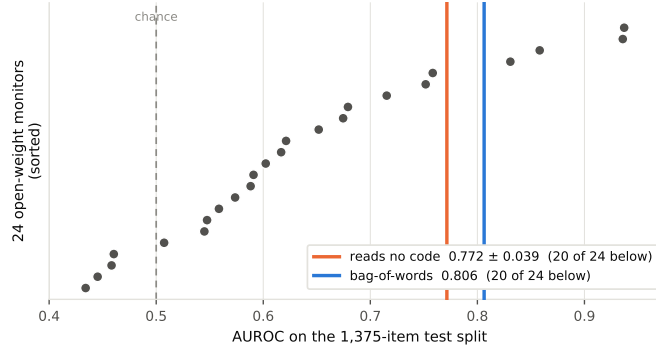


Figure 1: **Two floors against the 24 open-weight monitors ranked on the same split.** AUROC on Koran et al.’s 1,375-item test split (172 attack / 1,203 honest), comments stripped; monitor scores from [9] (MIT), median 0.609, max 0.937. **Two caveats.** (i) The monitors are zero-shot prompted while the bag-of-words classifier is *trained* on the benchmark’s own paired train split: a statement about what the corpus makes recoverable [6], not a claim that bag-of-words is a better monitor. (ii) The no-code floor is cross-validated *within* the test split (appendix A.2); on one held-out slice it is 0.716 (CI 0.6476–0.7796; $p = 0.030$), above 18 of 24, 12 with a paired interval excluding zero.

37 Much of the residue is authorship, not sabotage. Honest solutions come from APPS [7] and are
 38 human-written; every backdoor is model-written. The classifier transfers across backdoor generators
 39 and is nearly as strong on **inert** backdoors as on working ones — neither is a property of a backdoor.
 40 That bag-of-words separates human-written from model-written code is established out-of-field
 41 [14]; the known-ness of the mechanism is the argument, not a dent in it. The neighbouring field
 42 already applies the control missing here — CodeMirage gates its classes on line and character count
 43 [5], MultiAIGCD pairs to shared problem definitions [1] — so unmatched pairs are a defect by a
 44 published standard. Adjacent control work attacks other axes: [20] subverts monitors with injected
 45 text, [3] shows attacker-side selection inflates safety estimates, and [15] removes transcript length as
 46 a nuisance covariate; we publish one as a per-benchmark floor.

47 We contribute (i) a protocol $\times$ baseline measurement of what survives the field’s preprocessing;
 48 (ii) a same-generator control on two generators, attributing between a third and nine-tenths of the
 49 separability to authorship and most of the remainder to the attack prompt’s framing, with a negative
 50 result on the way: regenerating one half of a stored pair swaps the confound rather than removing
 51 it; (iii) a second selection artifact the filter installs by defining the negative class through attack
 52 failure; and (iv) a preprocessing checklist and a released split. **Scope:** this concerns monitor training,
 53 selection and benchmarking on *stored* pairs; our measurement is of one APPS-derived family plus
 54 five agentic settings. ControlArena’s live evaluations generate both classes and are not implicated.

55 2 Setup

56 **The confound is legible in the schema.** The canonical stored-pair artifact for APPS-based control
 57 evaluations [8] records `model` and `source` on every backdoor and no generator field on any honest
 58 solution (Table 2, appendix).

59 **Artifacts and provenance.** We use seven public corpora; their per-class provenance is the object of
 60 study: in the APPS family the honest class is human and the attack class is one of ten named models.
 61 **Six of the seven, from two publishers, declare no licence** [11, 17]; we report only aggregates over
 62 them and redistribute no corpus.

63 **Evaluation surface, and naming protocols.** Our headline protocol is the filtered, comment-
 64 stripped split of [12]: 1,209 train problems (2,418 paired items) and 1,375 test items (172 attack,
 65 1,203 honest), labelled by `backdoor_works`, reconstructed with the released code of [9]. Every
 66 comparison below names its split, metric and preprocessing. Generated code is scored by our own
 67 runner, applied to both classes (appendix A.11).

68 3 Results

69 The fix works, on the two carriers \$1 names (Table 3, appendix). What it does not reach, on that
70 same split, is a bag-of-words floor of 0.806 (95% CI 0.7704–0.8409 by bootstrap over problems;
71 permutation null 95th percentile 0.5465, $p < 0.005$), above 20 of the 24 monitors by point estimate
72 and 18 of them with a paired-bootstrap interval excluding zero, and a no-code floor of 0.772 (CI
73 0.7319–0.8005 on the pooled out-of-fold 0.766; null 0.5545, $p < 0.005$), above 20 and 17 respectively
74 (Figure 1): the preprocessing removed the carrier it targeted and not the two beside it.

75 **The signal is not sabotage.** Three measurements point the same way; every number here is
76 comment-stripped on both classes, the unstripped values, inflated by the comment channel, being in
77 appendix A.3. First, the classifier *transfers across backdoor generators* at 0.969 pairwise: trained
78 on one attacker’s backdoors it identifies another’s, so what it has learned is not any single model’s
79 sabotage style. Second, it is nearly as strong on **inert** backdoors (0.879) as on working ones (0.935);
80 if it were detecting sabotage, backdoors that do not function should be much harder. Third, the
81 residual carrier is stylistic rather than semantic: `__main__` appears in 0.116 of attack samples against
82 **0.000** of honest ones, replicated on the train split (152 versus 1 of 1,209) and untouched by the
83 stripper. None of these is a property of a backdoor, and a token-level test [14] does not replicate here
84 (appendix A.12).

85 **Regenerating one half of a pair does not isolate content.** Under the identical pipeline the human-
86 honest baseline is $B = 0.871$ over $n = 1,444$ pairs, and the routing rule and its bands were fixed
87 before any arm ran: retained signal $\rho = (G - 0.5)/(B - 0.5)$, with $\rho \leq 0.33$ reading as authorship
88 and $\rho > 0.72$ as content. With an open-weight model writing the honest class against the same
89 backdoors, separation *rises*, to $\rho = 1.234$ (appendix A.5): substituting the honest author replaced
90 one authorship confound with a larger one.

91 **One model writing both classes removes between a third and nine-tenths of the separability.**
92 We report the **independent** arm — the attack class written from the problem statement alone, which
93 changes authorship and nothing else, an **edit** arm shown its own honest solution returning near-copies
94 (appendix A.4). One outcome-blind draw per problem, the attack class as the benchmark defines it
95 — by intent plus passing the public tests (2.2% of these draws fire, the same-generator analogue of
96 the split’s own 80.7% inert class) — and B' inert-inclusive over the same problems (Table 1). The
97 first generator’s interval excludes the collapse band and spans the content line, so the pre-registered
98 rule returns *inconclusive between partial and content*; the second’s lies inside the collapse band. At
99 $n = 623$, B under the identical pipeline is 0.8521 ± 0.0055 , so the drop is not sample size.

Table 1: **Gate S: what is left when one model writes both classes.** S is the same-generator AUROC (5-fold, comments stripped) and B' the stored-pair baseline on the row’s own problems; ρ , its bands and its intervals are as in the text and appendix A.5. 97.75% of the first generator’s headline attacks never fire and 99.67% of the second’s; the strict rows are all-firing.

Arm	n	S (95% CI)	B'	ρ (95% CI, joint)	band
Sonnet 5, headline	623	0.751 (0.7295–0.7713)	0.8658	0.687 (0.625–0.755)	partial–content
Haiku 4.5, headline	611	0.5340 (0.5159–0.5531)	0.8584	0.095 (0.044–0.150)	collapse
Sonnet 5, prompt-matched	586	0.6423 (0.6179–0.6621)	0.8613	0.394 (0.326–0.454)	partial
Sonnet 5, strict	99	0.7652 (0.7016–0.8000)	0.7877	0.921 (0.662–1.089)	partial–content
Sonnet 5, strict, matched	96	0.7113 (0.6464–0.7453)	0.8030	0.697 (0.494–0.856)	partial–content

100 **Holding the attack prompt constant as well removes most of what is left.** What survives on
101 the first generator is lexical: length-only falls to 0.5851 while TF-IDF holds. It is not the trigger —
102 masking numerals and comparison operators leaves S unchanged, while normalising identifiers takes
103 it to 0.700 ($\rho = 0.54$) — and the top attack-side features are `def main` (67% vs 38.5% of sides) and
104 `data[`: the layout the attack prompt elicits. Regenerating the honest class under that framing minus
105 the flaw sentence takes ρ to 0.394, and the two honest classes alone, no attack on either side, separate
106 at 0.6006: what the prompt removes is framing, not backdoor. On the second generator neither moves
107 — 0.5399 with the prompt matched ($\rho = 0.113$) over a prompt-only floor of 0.4918 whose interval
108 contains chance: a generator that never adopts the prompt’s layout. The strict rows — attacks that
109 pass *and* fire under the benchmark’s own retry-and-keep, the rows that follow our Recommendation 3
110 — agree at low power, read through ρ alone because B at $n = 99$ is 0.7807 ± 0.0248 .

111 **Monitors are inflated the same way.** A zero-shot monitor prompt, answered by the model that wrote
 112 both classes, scores 0.9579 on the *stored* pairs over those problems and 0.566 (0.5507–0.5833) on the
 113 headline pairs, a monitor- ρ of 0.145 (0.111–0.182) against the classifier’s 0.687; on the strict pairs
 114 it is 0.9437 and 0.818 (0.7630–0.8686), monitor- ρ 0.717 (0.601–0.832), and a monitor that wrote
 115 neither class returns the same two ratios (0.151, a bound at half missingness, and 0.749). Classifier
 116 and monitor are independent pair by pair ($\phi = 0.04$). Our single-shot attacks fire given a pass at 2.5%
 117 against the stored attacker’s 67.2%: the stored attack class is a survivor set.

118 **The filter installs a second artifact.** Labelling by `backdoor_works` defines the negative class
 119 by *attack failure* at the problem level rather than by honesty: 80.7% of the test split’s labelled
 120 backdoors do not work (32.8% of those that pass the public tests), and a problem whose backdoor
 121 fails contributes its human honest solution to the negative class instead, so the label is fixed before
 122 any code is read. Attack problems also carry roughly half as many test cases (20.95 versus 38.85) —
 123 fewer tests, easier to backdoor undetectably — so difficulty and test count correlate with the label
 124 with no code read at all, which is where the content-blind 0.772 comes from.

125 **The negative control comes free.** In five same-generator agentic settings, where the same three
 126 models write both classes and no authorship confound is possible, the surviving floor is structural:
 127 0.504 (bash, on its null) to 0.802 (agentdojo) on character count alone (Figure 2, appendix).

128 4 Recommendations

129 Following Kapoor and Narayanan [10], who treat leakage as a checklist rather than a verdict:

- 130 1. **Generate both classes with the same model and the same prompt framing.** If the attack
 131 class is model-written, the honest class must be too; as our arm-1 null shows, regenerating
 132 only one side substitutes one authorship gap for another.
- 133 2. **Report a partial-input floor and a bag-of-words floor beside every monitor number,**
 134 each against a permutation null.
- 135 3. **Do not define the negative class by attack failure.** Drop the problems whose attacks failed,
 136 or sample the negative class independently of attack success.
- 137 4. **State the split, the metric and the preprocessing** whenever monitors are compared, and
 138 **release the split and the scripts.** Ours accompany this paper under MIT (appendix A.10).

139 5 Limitations

140 **A high floor is evidence about a corpus, not a verdict on results computed on it.** Feng et al. [2]
 141 warn that the *failure* of a partial-input baseline does not show a dataset is artifact-free; the converse
 142 binds us: our floors show the label is recoverable without reading a backdoor [6, 16], not that any
 143 monitor exploits that.

144 **The same-generator arm is a later model behind a paid API.** The stored attacker,
 145 `claude-3-7-sonnet-thinking`, is retired (404 against a live key), so both classes come from
 146 a later model of the same lineage, called through a metered commercial API that does not expose
 147 its sampling parameters: the arm is not bit-reproducible, and thinking was off although the stored
 148 attacker was a thinking model. We do not length-match; the untrusted model chooses its own trajec-
 149 tory length. The stored attacker and the same-generator attacker also differ in backdoor style, not
 150 only in authorship, so every ρ here — classifier or monitor — bounds how much of the separability
 151 is content and does not partition it; and the two generators are compared on the passes eligibility
 152 alone, Haiku 4.5’s strict row being refused at 35 usable pairs against our 50-pair floor.

153 **Scope, provenance and the harness.** Whether stored-pair corpora leak this way in general is not
 154 established here: five of our seven corpora are one user’s repackaging, and claims are scoped to those
 155 releases. Two interpreter-era artifacts fall on opposite classes; the same-generator arm is immune to
 156 them, B is not (appendix A.11).

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A Supplementary tables and figures

A.1 The schema

Table 2 is the record schema of `RoganInglis/apps-control-arena` [8]. The attack class carries `model` and `source`, naming the system that produced each backdoor; the honest class carries no such field, because there is no generator to name. The two APPS-backdoor corpora we use are two views of this one artifact: every problem in the filtered release [11] is in [8], byte-identical except a re-validated `backdoor_works`. Among the agentic corpora we exclude `AgentHarm`, whose author is an invited speaker at this venue.

Table 2: The confound is legible in the dataset schema, with no measurement at all. The attack class records which model wrote it; the honest class has no generator field. Verified against the shard schema of `RoganInglis/apps-control-arena`.

struct	sub-fields
<code>solutions</code>	<code>code</code> , <code>compiles</code> , <code>passes_tests</code>
<code>backdoors</code>	<code>backdoor_code</code> , <code>backdoor_input</code> , <code>backdoor_works</code> , <code>compiles</code> , <code>model</code> , <code>passes_tests</code> , <code>source</code>

A.2 The protocol $\times$ baseline table

The no-code floor is cross-validated *within* the test split because the released train/val split is paired and therefore metadata-balanced; its permutation null shuffles labels for fitting and scoring alike. Table 3 is the full measurement behind §3. Rows are not comparable across protocols; the preprocessing column says whether comments were stripped before the baseline read the code. Pairwise accuracies are given to three decimals because their resolution over 2,641 pairs is one pair. Intervals in §3 are bootstraps over problems (2,000 draws) and nulls are 200 label permutations; the held-out no-code row is barely above its own null (0.716 against a 95th percentile of 0.7023, $p = 0.030$), so the no-code claim rests on the cross-validated protocol.

A.3 Comment stripping on the matched-protocol numbers

The matched-pair and transfer measurements were first made with comments in. Every value in the authorship paragraph of §3 was re-measured with the same stripper applied to both classes, over five problem-disjoint splits (each transfer row is one fixed split). With comments kept the same quantities were transfer 0.994, working 0.979, inert 0.955, all pairs 0.962 and surface-feature AUROC 0.8306; stripped they are 0.969, 0.935, 0.879, 0.897 and 0.7259. Every one was inflated by the comment channel, which is this paper’s thesis applied to itself. The `#-presence` marker, 0.685 against 0.176 before stripping, closes to 0.007 against 0.008 after and is dropped; `__main__` is unchanged. One authorship leak survives the stripper because string literals are not comments: in the comment-stripped train split an author byline (Author :) survives in 23 of 1,209 honest samples against 2 attack samples, and a triple-quoted block in 43 against 9 — too rare to drive 0.806, but a literal record of authorship. Stripped, the matched-protocol TF-IDF AUROC is 0.8560, beside

Table 3: What the field’s preprocessing leaves behind. Every row is measured under the protocol named in column one; rows are not comparable across protocols. *Prep.* says whether comments were stripped before the baseline read the code; *no code* means the baseline never reads the program. Pairwise accuracies are given to three decimals, their resolution over the pair count.

Protocol	Baseline	Value	Prep.	Against
<i>AUROC</i>				
Koran’s exact split	TF-IDF word 1–2gram	0.8063	stripped	above 20 of 24
same	TF-IDF word 1–2gram	0.8914	kept	above 22 of 24
same, reads no code	difficulty + #tests + $ q $	0.7715 ± 0.0389	no code	above 20 of 24
same, held-out slice	difficulty + #tests + $ q $	0.7163	no code	above 18 of 24
same	code length only	0.4726	stripped	chance
same	#-count detector	0.4983	stripped	chance
same	#-count detector	0.6957	kept	—
Raw artifact, problem level	all metadata	0.8917	no code	—
same	difficulty only	0.6823	no code	—
same	<code>solution_passes_tests</code>	0.7881	no code	—
Matched pairs (P3)	all metadata	0.5000	no code	exactly chance by construction
<i>Pairwise accuracy</i>				
DSPy matched, problem-disjoint	TF-IDF word 1–2gram	0.897	stripped	tutorial monitor 0.69–0.81
working backdoors only	TF-IDF word 1–2gram	0.935	stripped	—
inert backdoors only	TF-IDF word 1–2gram	0.879	stripped	the signal is not sabotage
transfer, Control Tax → GPT-4-era set	TF-IDF word 1–2gram	0.969	stripped	—
all pairs, comments kept	TF-IDF word 1–2gram	0.962	kept	inflated by the comment channel
working, comments kept	TF-IDF word 1–2gram	0.979	kept	inflated by the comment channel
inert, comments kept	TF-IDF word 1–2gram	0.955	kept	inflated by the comment channel
transfer, comments kept	TF-IDF word 1–2gram	0.994	kept	inflated by the comment channel
same, raw artifact	code length only	0.826	kept	—

243 $B = 0.871$ and the Koran-split 0.806, all three on stripped code. The other cross-generator transfer
244 rows, stripped, are 0.866 (o3-mini to claude-3-7-sonnet, 82 test pairs), 0.844 (the reverse) and 0.764
245 (o1-mini to gemini-2.0-flash).

246 A.4 Pair similarity, and why the same-generator control needs two arms

247 An attack class built by editing its own honest solution changes pair similarity as well as authorship
248 (Table 4); the independent arm sits in the stored benchmark’s regime and if anything slightly further
249 from it, so it cannot manufacture a low S by making the classes alike.

Table 4: Character similarity of honest/attack pairs, comments stripped. The edit arm’s classes are near-copies; the independent arm reproduces the stored benchmark’s regime, where the two classes are largely independent programs.

Attack class	mean sim.	> 0.95	< 0.60	changes vs. the benchmark
stored benchmark	0.606	3.4%	45.4%	— (the regime to match)
edit (shown its own solution)	0.928	64.2%	4.0%	authorship <i>and</i> similarity
independent (problem only)	0.659	3.4%	34.8%	authorship only

250 A.5 Robustness of the same-generator rows

251 Intervals in the Gate S paragraph of §3 are bootstraps over problems, resampling S and B' jointly for
252 ρ ; this replaced a ± 2 fold-sd rule before the retry pool was read. Every ρ point we quote is the ratio
253 of the fold-mean AUROCs and every joint interval beside it is the bootstrap of the pooled out-of-fold

ρ , which is the same quantity computed two ways: the pooled points are 0.86 (strict, the row whose interval Table 1 quotes), 0.390 (prompt-matched headline), 0.662 (prompt-matched strict) and 0.097 (the second generator). Five fold seeds move the strict row’s ρ over 0.85–0.92. Retention is not the artifact: survivors and non-survivors separate at only 0.62, and S with non-firing attempts on the same problems is 0.7595 against 0.7687. The edit arm bounds S under near-copies (similarity 0.928, Table 4): 0.6266 over 427 pairs, $\rho = 0.371$. In arm 1, where only the honest half was regenerated, $G = 0.962$ against $B' = 0.874$ on that arm’s $n = 668$ eligible problems, and TF-IDF gains +0.088 while the length carrier collapses to 0.489. Our own attack prompt halves our single-shot firing rate: 52.0% of our test-passing draws offer a public test input as the trigger, which cannot diverge from the honest solution, and those fire 0.15% against 5.13% for novel triggers; no stored working backdoor has such a trigger (0 of 3,188). The rest is retry-and-keep.

265 A.6 The prompt-matched arm under the masks, and the second generator

266 Under prompt matching the identifier mask — the only mask of §3 that moved S — takes the headline
 267 row to 0.6264 ($\rho = 0.342$) where it takes the original honest class on the same 586 problems to
 268 0.7046: most of what identifier normalisation removed is what prompt matching removes. That
 269 masked row is the one that decides whether the residue is structure or naming, and under prompt
 270 matching it is band-ambiguous: its joint interval, 0.280–0.394, puts 40% of its mass in the collapse
 271 band. A character model (3–5-grams, same classifier and folds) retains more than the word tokeniser,
 272 0.6817 ($\rho = 0.493$), so the drop is not an artefact of tokenisation. On the strict pairs the matched row
 273 is $S = 0.7113$ ($n = 96$, $\rho = 0.697$, 0.494–0.856). The second generator’s pairs are more self-similar
 274 than the first’s (0.724 mean against 0.659), so S was recomputed inside similarity strata: it stays
 275 0.4737–0.5142 in every stratum, including the 0.451-similarity stratum, which is further from its
 276 honest class than the stored benchmark’s own pairs, while B' holds 0.7884–0.8334. The collapse is
 277 not bought with near-copies. Its prompt-matched rows move S to 0.5399 ($\rho = 0.113$, $n = 562$) and
 278 put the prompt-only floor at 0.4918, an interval containing chance. The mechanism is visible in the
 279 programs: the layout vocabulary the first generator’s residue is made of — `def main, sys.stdin,`
 280 `data[` — is present in 0.000 of the second generator’s programs in *all three* of its classes, which read
 281 standard input with `input(` in 1.000 of programs on every side, so there is no layout difference left
 282 for the classifier to read.

283 A.7 The monitor rows, and what bounds the second monitor

284 The monitor that wrote neither class scores 0.5577 on the same-generator headline pairs and 0.8824
 285 on the stored pairs over the same problems, 0.7871 and 0.8832 on the strict pairs. Its headline rows
 286 are a bound rather than a point: a 512-token reply ceiling left 628 of 1,246 same-generator replies
 287 unparseable, and on the stored side the missing share differs by class (50.1% honest against 23.4%
 288 attack), a bias of unknown sign. Re-running the strict pairs with a 1,024-token ceiling prices it:
 289 missingness falls to 2 of 198 and monitor- ρ moves from 0.749 to 0.766, inside the bootstrap interval.
 290 On the stored pairs both monitors beat the bag-of-words baseline over the same problems (+0.09 and
 291 +0.15 for the first, +0.02 and +0.09 for the second): Figure 1’s ranking is a statement about those
 292 monitors on that split, not a claim that a bag of words beats a monitor here.

293 A.8 The two generators’ funnel

294 Both generators were run through the identical pipeline. Our honest draws pass the public tests at
 295 0.7701 and 0.7278 of 1,444 problems; the second generator’s attack draws pass at 689 of 1,444 and
 296 35 of those fire, below the 50-pair floor `gate_s_samegen.py` requires, which is why it has no strict
 297 row and the two generators are compared on the *passes* eligibility only. The two compared rows
 298 are not equally inert either: 2 of the 611 attacks in the second generator’s row fire (99.67% inert)
 299 against 14 of 623 (97.75%), the other 33 of its firing draws sitting on problems where its own honest
 300 program failed the tests and which Gate S therefore drops. The single-shot firing rate is a property of
 301 the protocol rather than of either model: 52.0% of test-passing draws offer a public test input as their
 302 trigger.

303 A.9 The five same-generator agentic settings

304 Figure 2 shows each setting’s content-blind floor against its own permutation null.

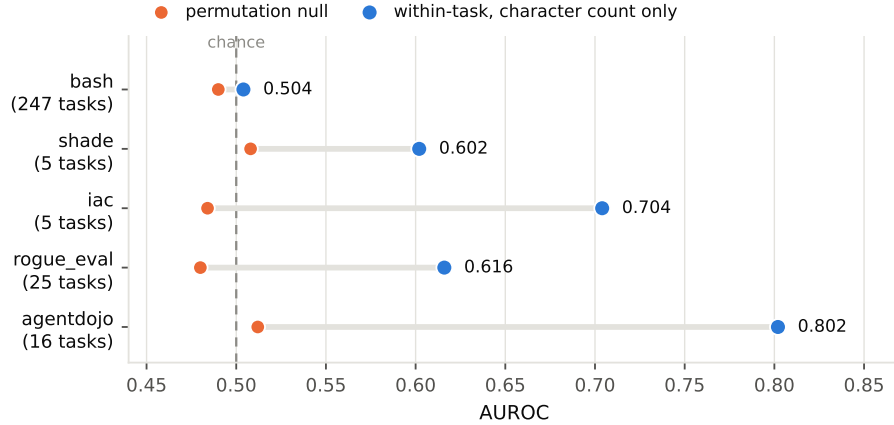


Figure 2: **The structural floor in five same-generator agentic settings.** Each setting’s content-blind classifier — six count features, no content read — scored *within task* against its own permutation null. No authorship confound is possible here: claude-sonnet-4, gemini-2.5-pro and gpt-5 write both classes in all five. bash sits on its null at 0.504; agentdojo reaches 0.802 from character count alone. **The task-group count beside each setting matters:** shade and iac average over five task groups each while bash averages over 247, so these floors are not equally well determined. We report the within-task column because our own pooled numbers partly broke on task composition, and we broke them ourselves. The permutation nulls shown were computed with a pooled estimator rather than the folded within-task statistic; they bound the chance level but are not a like-for-like null for each dot. These corpora declare no licence; we redistribute nothing and report aggregates only.

305 A.10 The release

306 The supplementary bundle holds the same-generator split (1,444 honest programs, 1,112 edit-arm
 307 and 1,112 independent-arm attack programs written by one model, plus every one of the 11,565
 308 draws of the independent arm’s retry pool), both eligibility rules as pair lists under an outcome-blind
 309 selection (623 passing and 14 firing independent-arm pairs) and under the benchmark’s own survivor
 310 selection (1,046 and 99, or 106 counting a crash on the trigger as firing), the 24 scripts behind the
 311 measurements — generation for both generators, the test harness, the trigger verifier, the selection
 312 rules, the prompt-matched arm, the monitor rows and each remaining measurement script — a dataset
 313 card, and a manifest of row counts and checksums. The licence, MIT, is a file inside the bundle and
 314 covers the data and the scripts alike.

315 A.11 The test harness and the pass flag

316 Gate S requires executing generated code, so we built a runner that feeds each shipped input case
 317 to a solution under per-case timeouts and address-space limits. It agrees with the artifact’s own
 318 `solution_passes_tests` on 95.84% of the analysis pool and is strictly conservative there — it
 319 never credits a solution the artifact rejects (0 such cases). We use *our own* flag for both classes:
 320 a checker stricter on one class than the other would manufacture exactly the artifact this paper
 321 documents. The headline does not depend on that choice — under the artifact’s flag $B = 0.871$ at
 322 $n = 1,444$, under ours $B = 0.873$ at $n = 1,406$, within one standard deviation of each other. The
 323 monitor scores in Figure 1 are runs/x0/monitor-summary.json of [9]’s MIT release, used under
 324 its notice. Two interpreter-era artifacts in the upstream solutions fall on opposite classes: a `stdin`
 325 without `.buffer`, fixed, and `fractions.gcd`, removed in Python 3.9, not fixable.

326 A.12 The token test, run and failed

327 We attempted to confirm the token-level mechanism of [14] in our setting and it does not repli-
 328 cate: separation -0.013 — the wrong sign — at permutation $p = 0.956$. The reason is register.
 329 Their human class is general-purpose repository code, whereas *both* of our classes are competitive-
 330 programming code, so an identifier axis that separates general human code from model code does not

331 separate these two populations. Only 8 of their 10 canonical tokens are in our vocabulary, so the test
332 is underpowered; but the point estimate has the wrong sign, which is not what an underpowered true
333 effect looks like. We report it because a reader who knows that work may run exactly this test.

334 **A.13 A third channel, noted rather than developed**

335 In the same five agentic settings a fifteen-word regular expression over the untrusted policy’s own
336 scratchpad reaches 0.76–0.82, because prompted attack policies restate their side task in the reasoning
337 the monitor is given. We note this only to bound the claim: the instrument belongs to [13], which
338 validates it properly against an LLM judge and a length-matched null, and we defer to that treatment.